

Worst-case error bounds for parallel machine scheduling problems with bounded sequence-dependent setup times

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We consider minimizing makespan ($C_{\max}$) and maximum lateness ($L_{\max}$) on parallel identical machines with sequence-dependent setup times. We show that the optimal schedule need not be a list schedule. Motivated by a practical scheduling problem, we study the class of problems where setup times are bounded by processing times and develop tight worst case error bounds for list scheduling algorithms to minimize $C_{\max}$ and $L_{\max}$.

scheduling; parallel machines; sequence-dependent setup times; heuristics

1. Introduction

We study the problems of minimizing makespan ($C_{\max}$) and maximum lateness ($L_{\max}$) on parallel identical machines with sequence-dependent setup times. We are given n jobs to be scheduled on m identical machines. Each job j has a processing time p_j and a due date d_j . A sequence-dependent setup time s_{ij} is incurred when job j is processed immediately following job i . All data is deterministic and known *a priori*. This problem is denoted by $P/s_{ij}/L_{\max}$ in Lageweg's notation [7]. We also study the special case of the problem where all due dates are identical, which we denote by $P/s_{ij}/C_{\max}$. These problems are motivated by the final testing stage of semiconductor manufacturing, where automated testing equipment is used to interrogate integrated circuits to determine whether they are operating to specification. Circuits must be tested at different temperatures, which leads to sequence-dependent setup times between different lots of circuits (jobs) due to the time taken to heat or cool the testing chamber to the required temperature.

When all due dates are identical the single machine case of this problem reduces to that of minimizing $C_{\max}$ on a single machine with sequence-dependent setup times, which is equivalent to the Travelling Salesman problem (TSP) [1] and strongly NP-hard. Sahni and Gonzalez [10] have shown that unless $P = NP$, no polynomial-time heuristic can yield a fixed data-independent worst-case error bound for the TSP with arbitrary intercity distances. However, in semiconductor testing the setup times are often bounded by the processing time of the lot to be tested. This motivates us to study problems where setup times are bounded by processing times, which have not received much attention in the literature to date. We derive tight data-independent worst case error bounds for list scheduling heuristics for this problem class. These results are of interest because they indicate that even when it is provably impossible for the general problem, exploitation of the structure present in practical scheduling problems may allow us to quantify the worst-case performance of heuristics.

2. Previous related work

List scheduling algorithms are a family of procedures for scheduling parallel identical machines that are based on some list or permutation π of the jobs. Whenever a machine becomes idle, the unscheduled job at the head of the list π is scheduled on that machine. Since the schedule generated by a list scheduling algorithm will differ according to the order in which the list π is ordered, we shall denote a general list scheduling algorithm by $\text{LIST}(\pi)$. $\text{LIST}(\text{Random})$ corresponds to the case where the jobs in π are sequenced in some random order, and $\text{LIST}(\text{EDD})$ to the case where they are in nondecreasing order of their due dates. It should be noted that list scheduling procedures generate non-delay schedules, where no machine is kept idle if there is a job available for processing.

Parallel machine scheduling problems without sequence-dependent setup times have been studied by several authors. Graham [6] analyzes the worst-case performance of arbitrary list scheduling algorithms and the $\text{LIST}(\text{LPT})$ procedure, where the jobs in the list are ordered in nonincreasing order of their processing times. Masuda et al. [9] address the problem of minimizing $L_{\max}$ and develop worst-case bounds on the performance of $\text{LIST}(\text{EDD})$ and a modified LPT list schedule. Carlier [3] gives a branch and bound procedure, and Carlier [3] and Gusfield [5] analyze the worst-case performance of the $\text{LIST}(\text{EDD})$ procedure for the $P/r_j/L_{\max}$ problem, where jobs become available dynamically over time.

Parallel machine scheduling problems with job families, where a major setup is incurred if the job scheduled is from a different family than the last job processed, have also been studied. Tang [11] presents a heuristic and two lower bounds on the optimal $C_{\max}$ for the problem where there are also minor setups between jobs of the same family. Bitran and Gilbert [2] develop a branch and bound procedure to assign jobs to machines and heuristics to schedule the jobs on each machine to minimize the number of major setups.

3. Non-dominance of list schedules

It can be shown that the family of list schedules is guaranteed to give the optimal schedule for static single machine problems with arbitrary setup times and regular performance measures and for static parallel machine problems without setups. In fact, this result can be extended to any problem where the set of non-delay schedules is dominant [4]. Given the optimal sequence to any instance of a problem, we can construct a list where jobs are ordered in increasing order of their start times in the optimal sequence. Applying the list scheduling procedure using this list will yield the optimal schedule.

Our first result shows that this is not possible for $P/s_{ij}/C_{\max}$, and hence for the more general $P/s_{ij}/L_{\max}$ problem. In the presence of sequence-dependent setup times, the set of list schedules is no longer dominant. In addition, there is no longer a one to one correspondence between a list and a schedule. These properties can be illustrated by a simple example with two machines and three jobs. The setup and processing times are given in Table 1 and the optimal solution in Fig. 1. The optimal schedule gives the list 1–2–3 or the list 2–1–3, neither of which corresponds to the optimal schedule. In both

Table 1
Processing and setup times for the list scheduling example

Job	p_i	SETUPS		To		
				1	2	3
1	1	From	0	1	1	10
2	2		1	0	10	10
3	3		2	10	0	1
			3	10	10	0

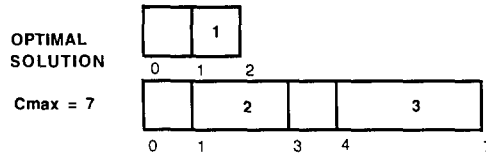


Fig. 1. Optimal schedule for list scheduling example

cases, the list scheduling algorithm schedules job 3 as soon as job 1 is completed. The better move is to delay job 3 for one time unit and process it after job 2, which no list schedule will do.

4. Worst-case bounds for problems with bounded setup times

For problems with arbitrary setup time values, the results of Sahni and Gonzalez [10] for the TSP imply that any polynomial-time procedure is very likely to have arbitrarily bad worst-case performance. In this section we show that if the setup times have some special structure, quantifying the worst-case performance of list schedules becomes possible. For the rest of this paper, we will assume that the setup time incurred for any job j is bounded by its processing time, i.e. $s_{ij} \leq p_j$ for all i . This assumption is valid for semiconductor testing operations, and is based on observation of the machinery in use.

We first develop a worst-case bound for $P/s_{ij}/C_{\max}$, which is a special case of $P/s_{ij}/L_{\max}$ where all release times and due dates are equal. We relate the worst-case performance of list scheduling procedures to the optimal $C_{\max}$ value of the problem without setup times, $P//C_{\max}$, which in turn is a lower bound on the optimal value of $P/s_{ij}/C_{\max}$.

Theorem 1. Let C^* denote the optimal value of the $P//C_{\max}$ problem, and $C_{\max}$ the makespan of the schedule obtained by any list scheduling algorithm for $P/s_{ij}/C_{\max}$. Assume also that $s_{ij} \leq p_j$ for all $i, j = 1, \dots, n$. Then

$$C_{\max} \leq \left(4 - \frac{2}{m}\right) C^*$$

and this bound is tight.

Proof. Clearly,

$$C^* \leq \sum p_i, \quad (1)$$

$$C^* \geq \max_i \{p_i\}, \quad (2)$$

$$C^* \geq \frac{1}{m} \sum p_i. \quad (3)$$

Let H_h be the time at which processing stops on machine h . Since the list scheduling algorithm leaves no machine idle while a job is available, we have $H_h \geq C_{\max} - s_{ik} - p_k$ where s_{ik} is the setup time and p_k the processing time of the last job completed. Since $s_{ij} \leq p_j$, we have

$$H_h \geq C_{\max} - 2p_k, \quad (4)$$

$$\sum_{h=1}^m H_h \leq 2 \sum_{i=1}^n p_i. \quad (5)$$

We assume, without loss of generality, that $C_{\max}$ is given by the last machine m , therefore $H_m = C_{\max}$. Summing this with (4) for all $h = 1, \dots, m-1$, and combining it with (5), we get

$$2 \sum_{i=1}^n p_i \geq \sum_{h=1}^m H_h \geq (m-1)(C_{\max} - 2p_k) + C_{\max}.$$

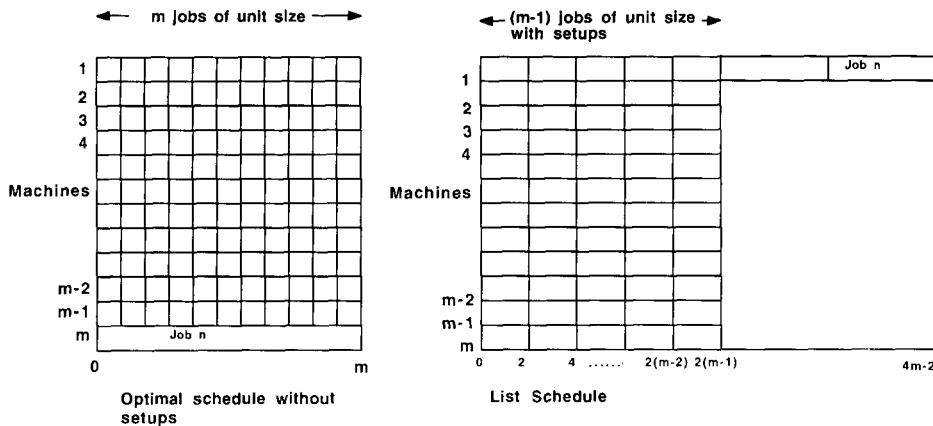


Fig. 2. Example for bound of Theorem 1

By (2) and (3), we have

$$2mC^* \geq mC_{\max} - 2(m-1)C^* \quad \text{since } p_k \leq C^*.$$

Rearranging the terms gives us

$$mC_{\max} \leq 2mC^* + 2(m-1)C^*,$$

which implies

$$C_{\max} \leq \frac{4m-2}{m}C^* = \left(4 - \frac{2}{m}\right)C^*. \quad \square$$

The following example shows that this bound is tight. For arbitrary m , consider $n = m(m-1) + 1$ jobs. Let jobs $1, 2, \dots, n-1$ have $p_i = 1$ and $s_{ij} = 1$ for $j = i+m, i = 1, \dots, m(m-2)$ and 0 otherwise. Let job n have $p_n = m$ and $s_{in} = m$ for $i = m(m-2) + 1$ and 0 otherwise. The optimal schedule without setups and the schedule obtained by the list scheduling algorithm using the list $1, 2, \dots, n$ are shown in Figure 2. The optimal schedule without setups has a makespan of m , while that of the list schedule is $4m-2$, showing that the bound of Theorem 1 is tight for arbitrary m .

We now use Theorem 1 to develop a worst-case bound for the LIST(EDD) procedure for the $P/s_{ij}/L_{\max}$ problem. To avoid problems caused by nonpositive $L_{\max}$, we convert the $L_{\max}$ values into makespan values for problems with delivery times as described by Lenstra [8]. This is done by adding a constant at least as large as the maximum due date to the $L_{\max}$ value. Hence the ratio

$$\frac{L - L^*}{L^* + d_{\max}}$$

represents the ratio of the deviation of the heuristic solution from the optimal solution to the optimal makespan of the problem with delivery times where $d_{\max}$ is the maximum due date of the jobs to be scheduled.

Theorem 2. Consider the LIST(EDD) algorithm for $P/s_{ij}/L_{\max}$. Let L^* denote the optimal value of $P//L_{\max}$, the problem without sequence-dependent setup times and L the value obtained by the LIST(EDD) algorithm. Then

$$\frac{L - L^*}{L^* + d_{\max}} \leq \left(3 - \frac{2}{m}\right),$$

if $s_{ij} \leq p_j$ for all i, j , and this bound is tight.

Proof. We first make some observations about the relationship between $P//C_{\max}$ and $P//L_{\max}$. Let J be the set of jobs to be scheduled, and J' any subset of J . Let $C^*(J')$ and $L^*(J')$ denote the optimal values of the $P//C_{\max}$, and $P//L_{\max}$ problems, respectively, where J' is the job set to be scheduled. We have

$$\begin{aligned} C^* &= C^*(J) \geq C^*(J'), \\ L^* &= L^*(J) \geq L^*(J'). \end{aligned} \quad (6)$$

This says that the optimal value of any subset of the jobs to be scheduled constitutes a lower bound on the optimal value of the original problem. This is the case since we cannot do any worse by scheduling a smaller set of jobs and the optimal schedule will contain no idle time.

Consider the schedule for $P/s_{ij}/L_{\max}$ formed by the LIST(EDD) procedure. Let h be the job that determines the $L_{\max}$ of the schedule, and set J' include job h and the set of jobs that are scheduled before job h . Clearly, once job h is scheduled, the jobs in the list after h do not change the $L_{\max}$ value.

Let $C(J')$ and $L(J')$ be the $C_{\max}$ and $L_{\max}$ of the schedules formed by LIST(EDD) when applied to job set J' . Similarly, let $d_{\max}(J')$ be the maximum due date of the jobs in set J' . Then,

$$L = L(J') = C_h - d_h \leq C(J') - d_h = C(J') - d_{\max}(J') \quad (7)$$

where C_i denotes the completion time of job i . The $L_{\max}$ of the schedule that LIST(EDD) generates using the entire job set is equal to the $L_{\max}$ of the schedule using only J' by the definition of J' . The inequality holds because the $C_{\max}$ of the schedule has to be greater than the completion time of any job in the schedule. The second equality is true because h is the last job scheduled in J' . Therefore $d_h = d_{\max}(J')$ since otherwise it would not be the last job scheduled in J' .

We also know that for job set J' ,

$$L^*(J') \geq C^*(J') - d_{\max}(J'). \quad (8)$$

The right hand side of this expression assumes that the job with the latest due date is scheduled last and is completed as soon as possible. This clearly is a lower bound on the optimal $L_{\max}$ of the schedule.

Note that all the optimal values that we have referred to so far, and will refer to from here on, are for the problem without setups, and all objective function values given by the list schedule are for the problems with sequence-dependent setup times. Now that we have a lower bound on the optimal $L_{\max}$ for the problem without setups and an upper bound on the value of the heuristic, we can express the ratio of the deviation of the heuristic solution from the optimal solution to the optimal $C_{\max}$ of the problem with delivery times by

$$\frac{L - L^*}{L^* + d_{\max}} \leq \frac{L(J') - L^*(J')}{L^*(J') + d_{\max}(J')}$$

due to $d_{\max}(J') \leq d_{\max}$, (6) and the first equation of (7). Substituting the inequality of (7) and (8) into this expression gives

$$\frac{L - L^*}{L^* + d_{\max}} \leq \frac{[C(J') - d_{\max}(J')] - [C^*(J') - d_{\max}(J')]}{C^*(J') - d_{\max}(J') + d_{\max}(J')}.$$

Rearranging, and cancelling the $d_{\max}(J')$ terms we get

$$\frac{L - L^*}{L^* + d_{\max}} \leq \frac{C(J') - C^*(J')}{C^*(J')} = \frac{C(J')}{C^*(J')} - 1.$$

The right hand side of this expression contains a term which is the ratio of the $C_{\max}$ of the heuristic schedule to the optimal $C_{\max}$ of the problem without setups, for which we have an upper bound due to Theorem 1. Therefore,

$$\frac{L - L^*}{L^* + d_{\max}} \leq \left(4 - \frac{2}{m}\right) - 1 = \left(3 - \frac{2}{m}\right). \quad \square$$

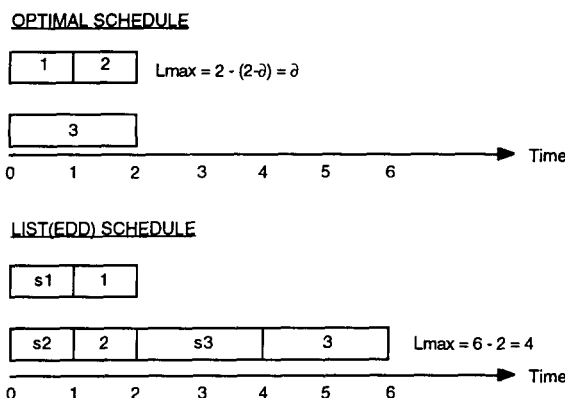


Fig. 3. Example for Theorem 2

The following example shows that this bound is tight. Let $m = 2$, and $n = 3$, $p_1 = p_2 = 1$, $p_3 = 2$, all setups $s_{ij} = p_j$ for $i > j$ and 0 otherwise, and $d_1 = 1$, $d_2 = 2 - \delta$, $d_3 = 2$ where δ is a small number. The optimal schedule for the problem without setups and the schedule for the problem with setups given by LIST(EDD) are shown in Figure 3. The ratio

$$\frac{L - L^*}{L^* + d_{\max}} = \frac{4 - \delta}{\delta + 2}$$

approaches the value 2 given by Theorem 2 as δ approaches 0, and therefore the bound is tight.

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